

Revision Notes for Class 8 Science Chapter 5 Exploring Forces

Every day, we experience pushes and pulls—these are known as forces. When you push or pull a box, pedal a bicycle, or open a door, you are applying a force. In science, a force is simply described as a push or a pull that results from the interaction between two objects. For example, when you push a table, your hand and the table are two different objects interacting with each other. The SI unit used to measure force is called newton (N).

What Can a Force Do?

A force can bring about different types of changes. It can start the motion of an object that was at rest, stop a moving object, change the speed of a moving object, or alter its direction. Sometimes, force can even change the shape of an object. For instance, pressing a balloon can change its shape, and hitting a moving ball with a bat changes its direction. A force may also cause more than one of these effects at the same time.

Interaction of Forces

A force always involves at least two objects—one that applies the force and one that receives it. No force can act on an object unless there is some interaction, either by touch or from a distance. Whenever we observe a change in speed, direction, or shape, it means a force is acting. When two objects interact, each experiences a force until the interaction ends.

Types of Forces: Contact and Non-contact

Forces can be divided into two main types: contact forces and non-contact forces. Contact forces only act when objects are physically touching, while non-contact forces can act even when objects are not touching.

Contact Forces

- **Muscular force:** This is the force generated by the action of muscles in humans and animals. We use muscular force for activities like walking, running, or lifting objects.
- **Friction:** Friction is the force that opposes the motion of one object moving or trying to move over another. It always acts in the opposite direction to movement. The rougher the surfaces in contact, the greater the friction. This is why it is easier to slip on smooth or wet surfaces and also why a ball rolling on sand stops faster than on a smooth floor.

Non-contact Forces

- **Magnetic force:** Magnets attract or repel without touching. Like poles of magnets repel each other, while unlike poles attract. This happens even if the magnets are not in direct contact.
- **Electrostatic force:** When certain materials are rubbed together, they become charged and can attract or repel other objects without contact. For example, a charged plastic scale can attract small pieces of paper.
- **Gravitational force:** This is the force by which the Earth pulls objects towards itself. It always acts downward, and does not need contact. Gravity is the reason why a thrown ball comes back to the ground, and why we stay anchored on Earth.

Friction and Its Importance

Friction results from the contact between two surfaces and always works against the direction of movement. All surfaces, even smooth ones, have tiny irregularities that lock together and create friction. The type of surfaces in contact affects how much friction there will be—rough surfaces cause more friction than smooth ones. Friction also acts in fluids (liquids and gases), which is why airplanes and ships are designed to minimize this force.

Weight and Its Measurement

Weight is defined as the force with which the Earth pulls an object toward itself. This force is measured in newton (N). To measure weight, we often use a spring balance, where the stretch in the spring indicates how much pull (force) is being exerted. The more an object weighs, the more it stretches the spring. The weight of any object may change depending on where it is—like on Earth, the Moon, or other planets—since the gravitational pull is different at different places.

Difference Between Mass and Weight

Mass refers to the amount of matter in an object and is measured in grams (g) or kilograms (kg). Unlike weight, mass does not change from place to place. Weight depends on both mass and the pull of gravity, so it can vary. For example, an object with 1 kg mass weighs about 10 N on Earth, 1.6 N on the Moon, and 25.4 N on Jupiter.

Weight of a 1 kg object on Different Planets

Planet	Weight
Earth	10 N
Moon	1.6 N
Mars	3.8 N
Venus	9 N
Jupiter	25.4 N

Buoyant Force and Archimedes' Principle

When an object is put into a liquid, it experiences an upward push known as upthrust or buoyant force. This force acts against gravity. According to Archimedes' Principle, a floating object experiences an upward force equal to the weight of the liquid it displaces. If this upward force equals or goes beyond the weight of the object, the object floats; if not, it sinks. The density of the liquid and the object decides whether it will float or sink.

Floating and Sinking

Objects like a wooden block float in water while a coin sinks because the upthrust is equal to or greater than the object's weight in some cases, but not in others. Some rocks like pumice can even float on water due to their low density and air pockets inside.

Key Snapshots

- Force is a push or pull due to interaction between objects, measured in newton (N).
- Friction and muscular force are examples of contact forces; magnetic, electrostatic, and gravitational forces are non-contact forces.
- Force can change an object's motion or shape.
- Friction always acts opposite to the direction of motion.
- Weight is based on gravity and can change depending on location, while mass remains constant.
- Objects float or sink in a liquid depending on the balance of their weight and the upthrust exerted by the liquid.